Unit II: Network Models & Services

1. Network Models – OSI/ISO Reference Model
2. TCP/IP Model
3. Protocol Hierarchies
4. Design issues for layers
5. Service Primitives - LISTEN, CONNECT, RECEIVE,SEND, DISCONNECT

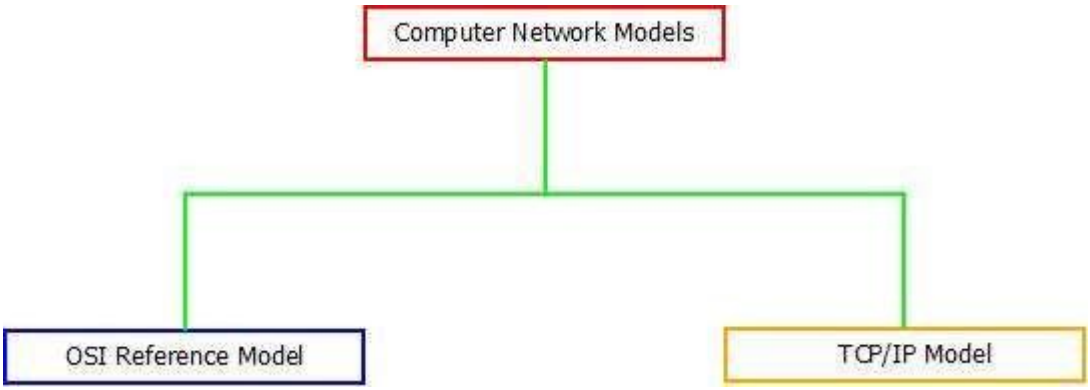
Computer Network Models

A computer network consists software and hardware that is used to send and receive data from one device to another. The role of hardware is to prove the physical equipment that are required in order to send and receive data while software defines the set of instructions that uses the hardware equipments for data transmission. A simple transmission of data consists several steps at various layers of computer network. In computer network models we will discuss the models in detail to understand how the data is actually transferred and received at a computer level.

Computer Network Models : For data communication to take place and two or more users can transmit data from one to other, a systematic approach is required. This approach enables users to communicate and transmit data through efficient and ordered path. It is implemented using models in computer networks and are known as computer network models.

Computer network models are responsible for establishing a connection among the sender and receiver and transmitting the data in a smooth manner respectively.

There are two computer network models i.e. OSI Model and TCP/IP Model on which the whole data communication process relies.



1.OSI Reference Model

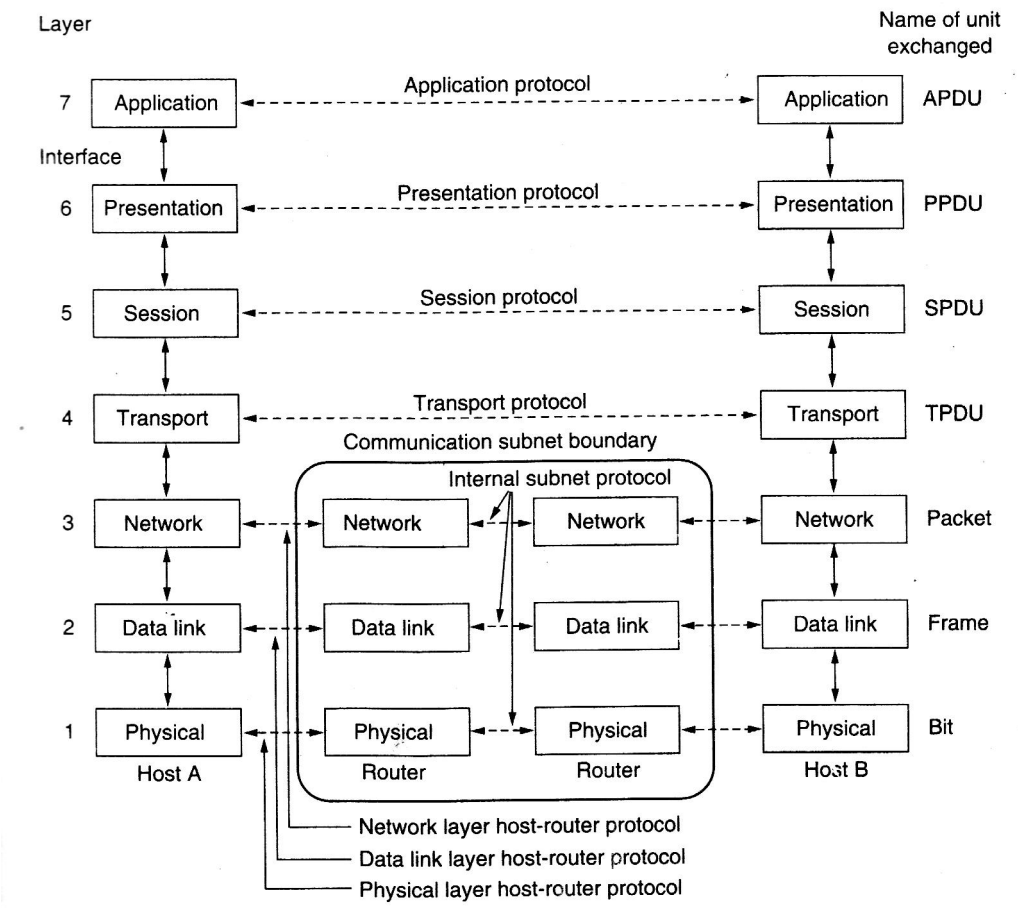
OSI stands for Open System Interconnection is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.

OSI consists of seven layers, and each layer performs a particular network function.

OSI model was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications.

OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned a particular task.

Each layer is self-contained, so that task assigned to each layer can be performed independently.



Functions of the OSI Layers

Physical layer

The main functionality of the physical layer is to transmit the individual bits from one node to another node.

It is the lowest layer of the OSI model.

It establishes, maintains and deactivates the physical connection.

It specifies the mechanical, electrical and procedural network interface specifications.

Functions of a Physical layer:

- Line Configuration: It defines the way how two or more devices can be connected physically.
- Data Transmission: It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.
- Topology: It defines the way how network devices are arranged.
- Signals: It determines the type of the signal used for transmitting the information.

Data-Link Layer

- This layer is responsible for the error-free transfer of data frames.
- It defines the format of the data on the network.
- It provides a reliable and efficient communication between two or more devices.
- It is mainly responsible for the unique identification of each device that resides on a local network.

It contains two sub-layers:

Functions of the Data-link layer

Framing: The data link layer translates the physical's raw bit stream into packets known as Frames. The Data link layer adds the header and trailer to the frame. The header which is added to the frame contains the hardware destination and source address.

Physical Addressing: The Data link layer adds a header to the frame that contains a destination address. The frame is transmitted to the destination address mentioned in the header.

Flow Control: Flow control is the main functionality of the Data-link layer. It is the technique through which the constant data rate is maintained on both the sides so that no data get corrupted. It ensures that the transmitting station such as a server with higher processing speed does not exceed the receiving station, with lower processing speed.

Error Control: Error control is achieved by adding a calculated value CRC (Cyclic Redundancy Check) that is placed to the Data link layer's trailer which is added to the message frame before it is sent to the physical layer. If any error seems to occur, then the receiver sends the acknowledgment for the retransmission of the corrupted frames.

Access Control: When two or more devices are connected to the same communication channel, then the data link layer protocols are used to determine which device has control over the link at a given time.

Network Layer

It is a layer 3 that manages device addressing, tracks the location of devices on the network.

It determines the best path to move data from source to the destination based on the network conditions, the priority of service, and other factors.

The Data link layer is responsible for routing and forwarding the packets.

Routers are the layer 3 devices; they are specified in this layer and used to provide the routing services within an internetwork.

The protocols used to route the network traffic are known as Network layer protocols. Examples of protocols are IP and Ipv6.

Functions of Network Layer:

Internetworking: An internetworking is the main responsibility of the network layer. It provides a logical connection between different devices.

Addressing: A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet.

Routing: Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.

Packetizing: A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).

Transport Layer

- The Transport layer is a Layer 4 ensures that messages are transmitted in the order in which they are sent and there is no duplication of data.
- The main responsibility of the transport layer is to transfer the data completely.
- It receives the data from the upper layer and converts them into smaller units known as segments.
- This layer can be termed as an end-to-end layer as it provides a point-to-point connection between source and destination to deliver the data reliably.

The two protocols used in this layer are:

Transmission Control Protocol

- It is a standard protocol that allows the systems to communicate over the internet.
- It establishes and maintains a connection between hosts.
- When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments. Each segment travels over the internet using multiple routes and they arrive in different orders at the destination. The transmission control protocol reorders the packets in the correct order at the receiving end.

User Datagram Protocol

User Datagram Protocol is a transport layer protocol.

It is an unreliable transport protocol as in this case receiver does not send any acknowledgment when the packet is received, the sender does not wait for any acknowledgment. Therefore, this makes a protocol unreliable.

Functions of Transport Layer:

Service-point addressing: Computers run several programs simultaneously due to this reason, the transmission of data from source to the destination not only from one computer to another computer but also from one process to another process. The transport layer adds the header that contains the address known as a service-point address or port address. The responsibility of the network layer is to transmit the data from one computer to another computer and the responsibility of the transport layer is to transmit the message to the correct process.

Segmentation and reassembly: When the transport layer receives the message from the upper layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segment. When the message has arrived at the destination, then the transport layer reassembles the message based on their sequence numbers.

Connection control: Transport layer provides two services Connection-oriented service and connectionless service. A connectionless service treats each segment as an individual packet, and they all travel in different routes to reach the destination. A connection-oriented service makes a connection with the transport layer at the destination machine before delivering the packets. In connection-oriented service, all the packets travel in the single route.

Flow control: The transport layer also responsible for flow control but it is performed end-to-end rather than across a single link.

Error control: The transport layer is also responsible for Error control. Error control is performed end-to-end rather than across the single link. The sender transport layer ensures that message reach at the destination without any error.

Session Layer

It is a layer 3 in the OSI model.

The Session layer is used to establish, maintain and synchronizes the interaction between communicating devices.

Functions of Session layer:

Dialog control: Session layer acts as a dialog controller that creates a dialog between two processes or we can say that it allows the communication between two processes which can be either half-duplex or full-duplex.

Synchronization: Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as

Synchronization and recovery.

- Presentation Layer It acts as a data translator for a network.
- This layer is a part of the operating system that converts the data from one presentation format to another format.
- The Presentation layer is also known as the syntax layer.

Functions of Presentation layer:

Translation: The processes in two systems exchange the information in the form of character strings, numbers and so on. Different computers use different encoding methods, the presentation layer handles the interoperability between the different encoding methods. It converts the data from sender-dependent format into a common format and changes the common format into receiver-dependent format at the receiving end.

Encryption: Encryption is needed to maintain privacy. Encryption is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.

Compression: Data compression is a process of compressing the data, i.e., it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

Application Layer

- An application layer serves as a window for users and application processes to access network service.
- It handles issues such as network transparency, resource allocation, etc.
- An application layer is not an application, but it performs the application layer functions.
- This layer provides the network services to the end-users.

Functions of Application layer:

File transfer, access, and management (FTAM): An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.

Mail services: An application layer provides the facility for email forwarding and storage.

Directory services: An application provides the distributed database sources and is used to provide that global information about various objects.

A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.

2.TCP/IP model

- The TCP/IP model was developed prior to the OSI model.
- The TCP/IP model is not exactly similar to the OSI model.
- The TCP/IP model consists of four layers: the application layer, transport layer, network layer, Network Access Layer
- The first four layers provide physical standards, network interface, internetworking, and transport functions that correspond to the first four layers of the OSI model and these four layers are represented in TCP/IP model by a single layer called the application layer.
- TCP/IP is a hierarchical protocol made up of interactive modules, and each of them provides specific functionality.
- Here, hierarchical means that each upper-layer protocol is supported by two or more lower-level protocols.

Functions of TCP/IP layers:

TCP/IP model

1.Network Access Layer

1. A network layer is the lowest layer of the TCP/IP model.
2. A network layer is the combination of the Physical layer and Data Link layer defined in the OSI reference model.
3. It defines how the data should be sent physically through the network.
4. This layer is mainly responsible for the transmission of the data between two devices on the same network.
5. The functions carried out by this layer are encapsulating the IP datagram into frames transmitted by the network and mapping of IP addresses into physical addresses.
6. The protocols used by this layer are ethernet, token ring, FDDI, X.25, frame relay.

2.Internet Layer

1. An internet layer is the second layer of the TCP/IP model.
2. An internet layer is also known as the network layer.
3. The main responsibility of the internet layer is to send the packets from any network, and they arrive at the destination irrespective of the route they take.

IP Addressing: This protocol implements logical host addresses known as IP addresses. The IP addresses are used by the internet and higher layers to identify the device and to provide internetwork routing.

Host-to-host communication: It determines the path through which the data is to be transmitted.

Routing: When IP datagram is sent over the same local network such as LAN, MAN, WAN, it is known as direct delivery. When source and destination are on the distant network, then the IP datagram is sent indirectly. This can be accomplished by routing the IP datagram through various devices such as routers.

3.Transport Layer

1. The transport layer is responsible for the reliability, flow control, and correction of data which is being sent over the network.

The two protocols used in the transport layer are User Datagram protocol and Transmission control protocol.

i) User Datagram Protocol (UDP)

a. It provides connectionless service and end-to-end delivery of transmission.
b. It is an unreliable protocol as it discovers the errors but not specify the error.
c. User Datagram Protocol discovers the error, and ICMP protocol reports the error to the sender that user datagram has been damaged.

UDP consists of the following fields:

i. Source port address: The source port address is the address of the application program that has created the message.
ii. Destination port address: The destination port address is the address of the application program that receives the message.
iii. Total length: It defines the total number of bytes of the user datagram in bytes.
iv. Checksum: The checksum is a 16-bit field used in error detection.

UDP does not specify which packet is lost. UDP contains only checksum; it does not contain any ID of a data segment.

Transmission Control Protocol (TCP)

1. It provides a full transport layer services to applications.
2. It creates a virtual circuit between the sender and receiver, and it is active for the duration of the transmission.

3. TCP is a reliable protocol as it detects the error and retransmits the damaged frames. Therefore, it ensures all the segments must be received and acknowledged before the transmission is considered to be completed and a virtual circuit is discarded.
4. At the sending end, TCP divides the whole message into smaller units known as segment, and each segment contains a sequence number which is required for reordering the frames to form an original message.
5. At the receiving end, TCP collects all the segments and reorders them based on sequence numbers.

4.Application Layer

1. An application layer is the topmost layer in the TCP/IP model.
2. It is responsible for handling high-level protocols, issues of representation.

3. This layer allows the user to interact with the application.
4. When one application layer protocol wants to communicate with another application layer, it forwards its data to the transport layer.
5. There is an ambiguity occurs in the application layer. Every application cannot be placed inside the application layer except those who interact with the communication system. For example: text editor cannot be considered in application layer while web browser using HTTP protocol to interact with the network where HTTP protocol is an application layer protocol.

Following are the main protocols used in the application layer:

i. HTTP: HTTP stands for Hypertext transfer protocol. This protocol allows us to access the data over the world wide web. It transfers the data in the form of plain text, audio, video. It is known as a Hypertext transfer protocol as it has the efficiency to use in a hypertext environment where there are rapid jumps from one document to another.
ii. SNMP: SNMP stands for Simple Network Management Protocol. It is a framework used for managing the devices on the internet by using the TCP/IP protocol suite.
iii. SMTP: SMTP stands for Simple mail transfer protocol. The TCP/IP protocol that supports the e-mail is known as a Simple mail transfer protocol. This protocol is used to send the data to another e-mail address.
iv. DNS: DNS stands for Domain Name System. An IP address is used to identify the connection of a host to the internet uniquely. But, people prefer to use the names instead of addresses. Therefore, the system that maps the name to the address is known as Domain Name System.
v. TELNET: It is an abbreviation for Terminal Network. It establishes the connection between the local computer and remote computer in such a way that the local terminal appears to be a terminal at the remote system.
vi. FTP: FTP stands for File Transfer Protocol. FTP is a standard internet protocol used for transmitting the files from one computer to another computer.

3.Protocol and Protocol Hierarchies

A protocol is a set of rules and conventions agreed upon and followed by the communicating entities for data communication. A protocol outlines the what, how and when of a communication.

The three aspects of a protocol are −

1) Syntax − It defines the format of data that is to be sent or received.

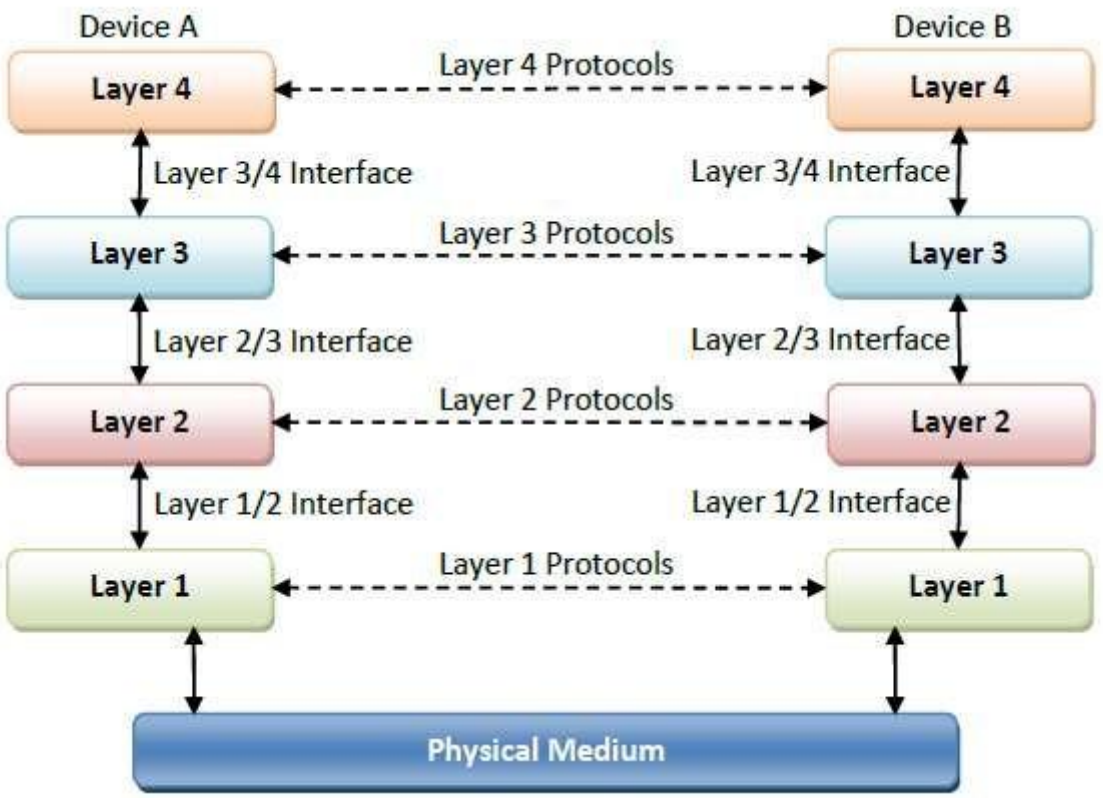
2) Semantics − It defines the meaning of each section of bits that are transferred.

3) Timings − It defines the time at which data is transferred as well as the speed at which it is transferred.

Protocol Hierarchies

Most networks are organized as a stack of layers, one on the top of another. The number of layers and their names vary from network to network. Each layer has a specified function and

adheres to specified protocols. Thus we obtain a stack of protocols.



- The above figure represents communication between Device A and Device B.
- The data stream from one device to the other is not sent directly but has to pass through a number of layers.
- The layers in the same levels are called peers and have a set of protocols for communication.
- Between each adjacent layer is an interface that defines the services that are being offered by a lower layer to the next higher layer.
- The dotted arrows depict virtual communication between peer layers, while the solid arrows represent the physical communications between the adjacent layers.

- Let us consider a situation where Device A wants to send a message to Device B.

- Device A passes its information to the highest layer.

- As soon as a data stream reaches a layer, it performs some specified functions on it and passes it to the layer below.

- This continues until the data stream reaches the lowest layer.
- Layer 1 passes a bit stream of 0s and 1s to the physical medium that communicates it to the Layer 1 of the receiving end.
- Each layer in the receiving end performs certain functions on the data stream adhering to the protocol with its peer and passes it to the layer above.
- This continues until the information reaches the highest layer. The highest layer then conveys the message to Device B in the same format sent by Device A.

Advantages :

- The layers generally reduce complexity of communication between networks
  It increases network lifetime.
- It also uses energy efficiently.
- It does not require overall knowledge and understanding of network.

4.Design Issues for the Layers of Computer Networks

A number of design issues exist for the layer to layer approach of computer networks. Some of the main design issues are as follows −

Reliability :
Network channels and components may be unreliable, resulting in loss of bits while data transfer. So, an important design issue is to make sure that the information transferred is not distorted.

Scalability :
Networks are continuously evolving. The sizes are continually increasing leading to congestion. Also, when new technologies are applied to the added components, it may

lead to incompatibility issues. Hence, the design should be done so that the networks are scalable and can accommodate such additions and alterations.

Addressing :

At a particular time, innumerable messages are being transferred between large numbers of computers. So, a naming or addressing system should exist so that each layer can identify the sender and receivers of each message.

Error Control:

Unreliable channels introduce a number of errors in the data streams that are communicated. So, the layers need to agree upon common error detection and error correction methods so as to protect data packets while they are transferred.

Flow Control:

If the rate at which data is produced by the sender is higher than the rate at which data is received by the receiver, there are chances of overflowing the receiver. So, a proper flow control mechanism needs to be implemented.

Resource Allocation :

Computer networks provide services in the form of network resources to the end users. The main design issue is to allocate and deallocate resources to processes. The allocation/deallocation should occur so that minimal interference among the hosts occurs and there is optimal usage of the resources.

Routing:

There may be multiple paths from the source to the destination. Routing involves choosing an optimal path among all possible paths, in terms of cost and time. There are several routing algorithms that are used in network systems.

Security:

A major factor of data communication is to defend it against threats like eavesdropping and surreptitious alteration of messages. So, there should be adequate mechanisms to prevent unauthorized access to data through authentication and cryptography.

5.Service Primitives

ii)Service generally includes set of various primitives. A primitive simply means Operations.

iii)A Service is specified by set of primitives that are available and given to user or other various entities to access the service.

iv) All these primitives simply tell the service to perform some action or to report on action that is taken by peer entity.

v) Each of the protocol that communicates in layered architecture also communicates in peer-to-peer manner with some of its remote protocol entity.

vi)Primitives are called calling functions between the layers that are used to manage communication among the adjacent protocol layers i.e., among the same communication node.

vii)The set of primitives that are available generally depends upon the nature of the service that is being provided.

Classification of Service Primitives :

1) **LISTEN** : When a server is ready to accept an incoming connection it executes the LISTEN primitive. It blocks waiting for an incoming connection.

2) **CONNECT** : It connects the server by establishing a connection. Response is awaited.

3) **RECIEVE**: Then the RECIEVE call blocks the server.

4) **SEND** : Then the client executes SEND primitive to transmit its request followed by the execution of RECIEVE to get the reply. Send the message.

5) **DISCONNECT** : This primitive is used for terminating the connection. After this primitive one can't send any message. When the client sends DISCONNECT packet then the server also sends the DISCONNECT packet to acknowledge the client. When the server package is received by client then the process is terminated.

Connection Oriented Service Primitives

There are 5 types of primitives for Connection Oriented Service :

LISTEN	Block waiting for an incoming connection
CONNECTION	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Sending message to the peer
DISCONNECT	Terminate a connection